Locality emerges from hidden zeros in $\mathrm{tr}(\phi^3)$: a by-hand proof at four, five, and six points

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Abstract

We prove that the cyclic *hidden one-zeros* of color-ordered $\mathrm{tr}(\phi^3)$ tree amplitudes force every non-local coefficient of the general rational ansatz to vanish, at $n = 4, 5, 6$ external legs. The proof is entirely by hand: we use only the kinematic mesh identity, geometric-series Laurent expansions in a distinguished "special variable," and the fact that a rational function of independent variables vanishes iff each monomial coefficient on a basis vanishes. The mechanism is geometric: each cyclic zero kills exactly the non-local terms whose chord factors involve the *enclosed vertex* of a designated short diagonal. Every non-local coefficient is killed by at least one zero. The four-point case is a one-line warm-up that delivers unitarity (equal residues), not locality. The five-point case kills all ten non-local coefficients of a fifteen-parameter ansatz. The six-point case extends the technique to three Laurent layers and kills all one-hundred-fifty-one non-local coefficients of a one-hundred-sixty-five-parameter ansatz, leaving exactly the fourteen triangulations.

Contents

1 Setup **2**
- 1.1 Chords, the ansatz, and locality . . . 2
- 1.2 The kinematic mesh identity . . . 2
- 1.3 The hidden one-zeros . . . 2
- 1.4 The geometric heart: enclosed vertices . . . 3

2 Four points: trivial warm-up **3**

3 Five points: complete proof **4**
- 3.1 The pentagon's chords and crossings . . . 4
- 3.2 The 15-coefficient ansatz . . . 4
- 3.3 The five zeros, derived by hand . . . 5
- 3.4 The pattern: every zero is a cyclic rotation of the previous . . . 5
- 3.5 The kill technique, in full detail at $\mathcal{Z}_1$ . . . 5
- 3.6 Cycling: kills from $\mathcal{Z}_2, \ldots, \mathcal{Z}_5$ . . . 6
- 3.7 Union: every non-local coefficient is killed . . . 7
- 3.8 The geometric pattern revealed . . . 7

4 Six points: the hexagon case **7**
- 4.1 Setup at $n = 6$ . . . 8
- 4.2 The six zeros, derived by hand . . . 8
- 4.3 The kill technique at $\mathcal{Z}_1$: three layers . . . 8
- 4.4 Cycling: union over six zones is 151 . . . 10

5 Looking ahead: the structural pattern 10

1 Setup

1.1 Chords, the ansatz, and locality

Label the external legs of a color-ordered n-point $\mathrm{tr}(\phi^3)$ amplitude cyclically by $1, 2, \ldots, n$. The *planar Mandelstam variables* are

$$X_{ij} \;=\; (p_i + p_{i+1} + \cdots + p_{j-1})^2, \qquad 1 \le i < j \le n, \;\; j - i \ge 2, \;\; (i,j) \ne (1,n).$$

Geometrically X_{ij} is the chord from vertex i to vertex j of the convex n-gon. There are $n(n-3)/2$ such chords, so 2, 5, 9 for $n = 4, 5, 6$.

A *triangulation* of the n-gon is a maximal set of non-crossing chords; any triangulation has $n-3$ chords and carves the polygon into $n-2$ triangles. The tree amplitude is the sum over triangulations of the product of chord propagators,

$$A_n \;=\; \sum_{T \text{ triangulation}} \prod_{(i,j)\in T} \frac{1}{X_{ij}}. \tag{1}$$

The number of triangulations is the Catalan number C_{n-2}: 2, 5, 14 for $n = 4, 5, 6$.

We consider the most general rational function B with the correct mass dimension $-2(n-3)$ and only planar simple poles:

$$B \;=\; \sum_{\substack{\text{multisets} \\ (a_1,\ldots,a_{n-3})}} \frac{a_{a_1\cdots a_{n-3}}}{X_{a_1}\cdots X_{a_{n-3}}}. \tag{2}$$

The sum runs over multisets (unordered with repetition) of size $n-3$ drawn from the set of chord indices. There are $\binom{d+n-4}{n-3}$ such multisets, where $d = n(n-3)/2$ is the number of chords; concretely $2, 15, 165$ for $n = 4, 5, 6$.

A multiset is *local* if its chords form a triangulation (in particular all chords are distinct and pairwise non-crossing). The remaining multisets are *non-local* – either they repeat a chord (producing a double pole $1/X^2$) or they pair two crossing chords. *Locality* is the statement that all non-local coefficients vanish.

1.2 The kinematic mesh identity

The non-planar two-particle invariants $c_{a,b} = 2p_a \cdot p_b$ (for $a \ne b$, non-adjacent) are not independent of the planar variables. They satisfy the linear *mesh identity*

$$\boxed{c_{a,b} \;=\; X_{a,b+1} + X_{a+1,b} - X_{a,b} - X_{a+1,b+1}} \tag{3}$$

where we set any "adjacent-chord" $X_{a,a+1}$ equal to zero (the external legs are massless, $p_a^2 = 0$). Setting $c_{a,b} = 0$ imposes one linear relation on the planar chords.

1.3 The hidden one-zeros

For each external leg $r \in \{1, \ldots, n\}$, the *hidden one-zero* $\mathcal{Z}_r$ is the linear subspace of kinematic space defined by

$$\mathcal{Z}_r : \qquad c_{r,r+2} \;=\; c_{r,r+3} \;=\; \cdots \;=\; c_{r,r+n-2} \;=\; 0 \pmod{n}.$$

There are $n-3$ constraints, and n zones cyclically permuted by leg relabeling. The starting fact – proved elsewhere and taken here as given – is that the tree amplitude A_n vanishes identically on each $\mathcal{Z}_r$. Our problem is the *converse* for the general ansatz (2): assuming $B|_{\mathcal{Z}_r} = 0$ for all r, show that every non-local a-coefficient vanishes.

1.4 The geometric heart: enclosed vertices

Every chord X_{ij} divides the boundary of the n-gon into two arcs. The shorter arc – equivalently, the one with fewer vertices strictly between i and j – is the *enclosed arc*, and the vertices strictly between i and j on that arc form the *enclosed-vertex set*

$$E(X_{ij}) \;=\; \{\text{vertices strictly between } i \text{ and } j \text{ on the shorter arc}\}.$$

We call $|E(X_{ij})|$ the *class* of the chord: class-1 chords are *short diagonals*, class-2 chords are *long diagonals*, and so on. The pentagon has only class-1 chords; the hexagon has both class 1 and class 2.

Lemma 1 (Crossing = enclosed-vertex sweep)**.** *Two chords X_{ij} and X_{rs} of a convex polygon cross in the interior if and only if exactly one endpoint of X_{rs} lies in $E(X_{ij})$ (equivalently in the enclosed arc of X_{ij}), and the other lies in the outer arc.*

Proof. Two chords of a convex polygon cross iff their four endpoints alternate around the boundary. Fix X_{ij} and consider the two arcs it cuts off. The endpoints of X_{rs} either both lie in the same arc (no alternation, no crossing) or one in each (alternation, crossing). □

Consequence. Every non-local pair of chords – that is, every crossing pair – has the form "one chord with enclosed vertex v; another chord starting at v." The hidden zeros are tailor-built to kill such pairs, as we will see.

2 Four points: trivial warm-up

The square has just two chords, X_{13} and X_{24}, and the ansatz

$$B \;=\; \frac{a_{X_{13}}}{X_{13}} + \frac{a_{X_{24}}}{X_{24}}$$

has two coefficients, both *local* (each chord by itself is a triangulation). There are no non-local multisets to kill.

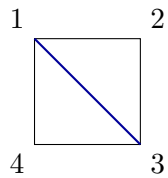
The single hidden zero $\mathcal{Z}_1$ imposes $c_{1,3} = 0$; the mesh identity gives

$$c_{1,3} \;=\; X_{1,4} + X_{2,3} - X_{1,3} - X_{2,4} \;=\; -X_{13} - X_{24} \;=\; 0 \quad\Longrightarrow\quad X_{24} = -X_{13}.$$

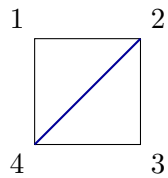
On this locus,

$$B\big|_{\mathcal{Z}_1} \;=\; \frac{a_{X_{13}}}{X_{13}} + \frac{a_{X_{24}}}{-X_{13}} \;=\; \frac{a_{X_{13}} - a_{X_{24}}}{X_{13}} \;=\; 0 \quad\Longrightarrow\quad \boxed{a_{X_{13}} \;=\; a_{X_{24}}.}$$

This is *unitarity* (equal residues on the two planar channels), not locality. The four-point zero has nothing to do with locality; the amplitude is already local by construction of the ansatz. The question becomes nontrivial at $n = 5$.



$1/X_{13}$



$1/X_{24}$

3 Five points: complete proof

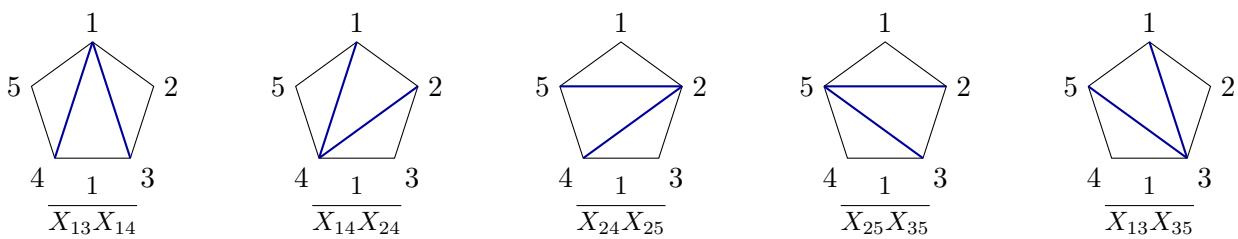
3.1 The pentagon's chords and crossings

The pentagon's five chords, listed with their enclosed-vertex sets, all have class 1:

Chord	X_{13}	X_{14}	X_{24}	X_{25}	X_{35}
$E(\cdot)$	$\{2\}$	$\{5\}$	$\{3\}$	$\{1\}$	$\{4\}$

Of the $\binom{5}{2} = 10$ unordered pairs of distinct chords, five share an endpoint (non-crossing, the five fan triangulations) and five cross. By Lemma 1 the crossing pairs are precisely those in which one chord's enclosed vertex is an endpoint of the other:

$$X_{13}\times X_{24}, \quad X_{13}\times X_{25}, \quad X_{14}\times X_{25}, \quad X_{14}\times X_{35}, \quad X_{24}\times X_{35}.$$



The five triangulations (fan triangulations of the pentagon).

3.2 The 15-coefficient ansatz

With $n - 3 = 2$ chord factors per monomial, the ansatz (2) reads

$$B \;=\; \sum_{1\le i\le j\le 5} \frac{a_{ij}}{X_i X_j}, \qquad 15 \text{ coefficients.}$$

Of these:

- 5 *tree triples* – one for each fan triangulation;
- 5 *double poles* $a_{X_{ij}X_{ij}}$ – one per chord;
- 5 *crossing pairs* $a_{X_{ij}X_{kl}}$ – one per crossing.

The latter ten are the non-local coefficients we must kill.

3.3 The five zeros, derived by hand

Each $\mathcal{Z}_r$ is $c_{r,r+2} = c_{r,r+3} = 0$ (two constraints; indices mod 5). Applying the mesh identity (3) and using $X_{a,a+1} = 0$ gives linear relations among the five chords.

$\mathcal{Z}_1$**:** $c_{1,3} = X_{14} - X_{13} - X_{24} = 0$ and $c_{1,4} = X_{24} - X_{14} - X_{25} = 0$. Solving,

$$\mathcal{Z}_1 : \qquad X_{24} \;=\; X_{14} - X_{13}, \qquad X_{25} \;=\; -X_{13}, \qquad \text{special } X_* \;=\; X_{13}. \tag{4}$$

The chord X_{13} appears on the right-hand side of *both* substitutions; we call it the *special variable* of $\mathcal{Z}_1$. The two substituted chords X_{24} and X_{25} are exactly the chords starting at the enclosed vertex $2 \in E(X_{13})$.

By cyclic symmetry, the remaining four zones are

Zone	Substitutions		Special X_*
$\mathcal{Z}_1$	$X_{24} = X_{14} - X_{13}$	$X_{25} = -X_{13}$	X_{13}
$\mathcal{Z}_2$	$X_{35} = X_{25} - X_{24}$	$X_{13} = -X_{24}$	X_{24}
$\mathcal{Z}_3$	$X_{14} = X_{13} - X_{35}$	$X_{24} = -X_{35}$	X_{35}
$\mathcal{Z}_4$	$X_{25} = X_{24} - X_{14}$	$X_{35} = -X_{14}$	X_{14}
$\mathcal{Z}_5$	$X_{13} = X_{35} - X_{25}$	$X_{14} = -X_{25}$	X_{25}

3.4 The pattern: every zero is a cyclic rotation of the previous

The structure of every $\mathcal{Z}_r$ is identical:

- *One special chord* X_* appears on the right of both substitutions.
- *Two substituted chords* – the chords from the enclosed vertex of X_*. One has the form $X_s = X_f - X_*$ with a non-zero *companion* X_f; the other is *bare*, $X_s = -X_*$.
- *Two free chords* remain. One of them is the companion of the non-bare substitution.

Write F_r for the free set. For $\mathcal{Z}_1$: $X_* = X_{13}$, companion $X_f = X_{14}$, bare substitution $X_{25} = -X_{13}$, free set $F_1 = \{X_{14}, X_{35}\}$.

3.5 The kill technique, in full detail at $\mathcal{Z}_1$

After the two substitutions, $B|_{\mathcal{Z}_1}$ is a rational function of three variables: the special X_{13} and the free X_{14}, X_{35}. The condition $B|_{\mathcal{Z}_1} = 0$ means this rational function vanishes identically. We extract single-coefficient kills by Laurent-expanding in the special variable X_{13}, treating X_{14} and X_{35} as fixed, and reading off coefficients at each order in $1/X_{13}$.

Step 1: Order X_{13}^0 (limit $X_{13} \to \infty$). As $X_{13} \to \infty$ with X_{14}, X_{35} held fixed:

$$X_{24} \;=\; X_{14} - X_{13} \;\to\; -\infty, \qquad X_{25} \;=\; -X_{13} \;\to\; -\infty,$$

so $1/X_{24} \to 0$ and $1/X_{25} \to 0$. Every monomial of B containing any factor of X_{13}, X_{24}, or X_{25} vanishes in the limit. The surviving monomials use only the two free chords, X_{14} and X_{35}:

$$\lim_{X_{13}\to\infty} B\big|_{\mathcal{Z}_1} \;=\; \frac{a_{X_{14}X_{14}}}{X_{14}^2} + \frac{a_{X_{35}X_{35}}}{X_{35}^2} + \frac{a_{X_{14}X_{35}}}{X_{14}X_{35}}.$$

This must vanish identically as a function of two independent variables X_{14}, X_{35}. The three rational functions $1/X_{14}^2, 1/X_{35}^2, 1/(X_{14}X_{35})$ are linearly independent, so each coefficient separately:

$$\boxed{a_{X_{14}X_{14}} \;=\; a_{X_{35}X_{35}} \;=\; a_{X_{14}X_{35}} \;=\; 0. \qquad (\mathcal{Z}_1,\ \text{three order-0 kills})}$$

Step 2: Order X_{13}^{-1}. Expand the substituted chords as geometric series:

$$\frac{1}{X_{24}} = \frac{1}{X_{14}-X_{13}} = -\frac{1}{X_{13}}\cdot\frac{1}{1-X_{14}/X_{13}} = -\frac{1}{X_{13}} - \frac{X_{14}}{X_{13}^2} - \frac{X_{14}^2}{X_{13}^3} - \cdots,$$

$$\frac{1}{X_{25}} = -\frac{1}{X_{13}} \qquad \text{(no tail)}.$$

Tracking the $1/X_{13}$ coefficient of each monomial gives

$$\left(B\big|_{\mathcal{Z}_1}\right)_{1/X_{13}} = \frac{a_{X_{13}X_{14}} - a_{X_{24}X_{14}} - a_{X_{25}X_{14}}}{X_{14}} + \frac{a_{X_{13}X_{35}} - a_{X_{24}X_{35}} - a_{X_{25}X_{35}}}{X_{35}}.$$

Setting this to zero and using linear independence of $1/X_{14}$ and $1/X_{35}$:

$$a_{X_{13}X_{14}} = a_{X_{24}X_{14}} + a_{X_{25}X_{14}}, \qquad a_{X_{13}X_{35}} = a_{X_{24}X_{35}} + a_{X_{25}X_{35}}.$$

These are *multi-term* relations, not single-coefficient kills.

Step 3: Order X_{13}^{-2}. Tabulate monomial-by-monomial. The crucial new piece is the crossing-pair monomial $a_{X_{24}X_{35}}/(X_{24}X_{35})$: its $1/X_{13}^2$ contribution is

$$-\frac{a_{X_{24}X_{35}}\, X_{14}}{X_{13}^2\, X_{35}},$$

producing a free-variable structure X_{14}/X_{35}. *No other monomial of B contributes X_{14}/X_{35} at order* X_{13}^{-2}: the other monomials at this order either evaluate to pure constants (from the leading $-1/X_{13}$ piece of a substituted chord inside monomials like $1/(X_{13}X_{24}), 1/X_{24}^2$, etc.) or produce different free-variable monomials. The coefficient of X_{14}/X_{35} at $1/X_{13}^2$ must therefore vanish:

$$\boxed{a_{X_{24}X_{35}} = 0. \qquad (\mathcal{Z}_1,\ \text{single order-2 kill})}$$

(The constant part of $(B|_{\mathcal{Z}_1})_{1/X_{13}^2}$ gives a multi-term equation involving seven coefficients, which we don't pursue here.)

Step 4: Orders $X_{13}^{-3}, X_{13}^{-4}, \ldots$. Direct check: no new single-coefficient kills arise. The same monomial X_{14}/X_{35} recurs (giving $a_{X_{24}X_{35}} = 0$ redundantly), and new monomials produce multi-term equations.

Summary of $\mathcal{Z}_1$.

$$\text{single-coefficient kills:} \quad \underbrace{a_{X_{14}X_{14}},\ a_{X_{35}X_{35}},\ a_{X_{14}X_{35}}}_{\text{order 0}},\ \underbrace{a_{X_{24}X_{35}}}_{\text{order 2}}. \qquad \text{(4 kills)}$$

3.6 Cycling: kills from $\mathcal{Z}_2, \ldots, \mathcal{Z}_5$

By cyclic symmetry each $\mathcal{Z}_r$ has the same internal structure, rotated:

Zone	Order-0 kills (multisets in F_r)			Order-2 kill
$\mathcal{Z}_1$ ($X_* = X_{13}$)	$a_{X_{14}X_{14}}$	$a_{X_{35}X_{35}}$	$a_{X_{14}X_{35}}$	$a_{X_{24}X_{35}}$
$\mathcal{Z}_2$ ($X_* = X_{24}$)	$a_{X_{14}X_{14}}$	$a_{X_{25}X_{25}}$	$a_{X_{14}X_{25}}$	$a_{X_{14}X_{35}}$
$\mathcal{Z}_3$ ($X_* = X_{35}$)	$a_{X_{13}X_{13}}$	$a_{X_{25}X_{25}}$	$a_{X_{13}X_{25}}$	$a_{X_{14}X_{25}}$
$\mathcal{Z}_4$ ($X_* = X_{14}$)	$a_{X_{13}X_{13}}$	$a_{X_{24}X_{24}}$	$a_{X_{13}X_{24}}$	$a_{X_{13}X_{25}}$
$\mathcal{Z}_5$ ($X_* = X_{25}$)	$a_{X_{24}X_{24}}$	$a_{X_{35}X_{35}}$	$a_{X_{24}X_{35}}$	$a_{X_{13}X_{24}}$

3.7 Union: every non-local coefficient is killed

Double poles (5 total). Each is killed at order 0 by exactly two zones:

$$\begin{array}{ll} a_{X_{13}^2}\colon \mathcal{Z}_3, \mathcal{Z}_4 & a_{X_{14}^2}\colon \mathcal{Z}_1, \mathcal{Z}_2 \\ a_{X_{24}^2}\colon \mathcal{Z}_4, \mathcal{Z}_5 & a_{X_{25}^2}\colon \mathcal{Z}_2, \mathcal{Z}_3 \\ a_{X_{35}^2}\colon \mathcal{Z}_1, \mathcal{Z}_5 & \end{array}$$

Crossing pairs (5 total). Each is killed once at order 0 and once at order 2:

$$\begin{array}{ll} a_{X_{13}X_{24}}\colon & \text{order-0 by } \mathcal{Z}_4,\ \text{order-2 by } \mathcal{Z}_5 \\ a_{X_{13}X_{25}}\colon & \text{order-0 by } \mathcal{Z}_3,\ \text{order-2 by } \mathcal{Z}_4 \\ a_{X_{14}X_{25}}\colon & \text{order-0 by } \mathcal{Z}_2,\ \text{order-2 by } \mathcal{Z}_3 \\ a_{X_{14}X_{35}}\colon & \text{order-0 by } \mathcal{Z}_1,\ \text{order-2 by } \mathcal{Z}_2 \\ a_{X_{24}X_{35}}\colon & \text{order-0 by } \mathcal{Z}_5,\ \text{order-2 by } \mathcal{Z}_1 \end{array}$$

Tree triples (5 total) are never killed. For each fan triangulation's coefficient $a_{X_{ij}X_{jk}}$, at every zone $\mathcal{Z}_r$ at least one of the two chord indices is either the special X_*, the bare substituted chord, or a substituted chord positioned so that it breaks the conditions for both the order-0 and order-2 kill. Equivalently: the tree amplitude A_5 vanishes on each $\mathcal{Z}_r$, so every Laurent coefficient of $A_5|_{\mathcal{Z}_r}$ vanishes; the tree triples sit inside the *kernel* of all our conditions.

Theorem 1 (Locality at five points)**.** *The five hidden one-zeros of the pentagon force every non-local coefficient of the general rational ansatz* (2) *to vanish. The surviving* 5 *coefficients are precisely the tree triangulations.*

Proof. By the explicit calculation at $\mathcal{Z}_1$ (Steps 1–4) and cyclic rotation to obtain $\mathcal{Z}_2, \ldots, \mathcal{Z}_5$, the union of single-coefficient kills covers all 10 non-local coefficients (5 double poles +5 crossing pairs), as tabulated. The 5 tree triples are never killed. □

3.8 The geometric pattern revealed

Every crossing pair is killed by *exactly two* zones, and those zones are *adjacent* in the cyclic ordering. The order-0 kill comes from the zone in which both chords of the crossing pair lie in the free set; the order-2 kill comes from the next zone over, where the special variable becomes one of the chord-pair's neighbors via the enclosed-vertex correspondence.

This is the **enclosed-vertex correspondence in action**: hidden zeros are tightly linked to crossings of the polygon, and every crossing has a precise pair of killing zones determined by the enclosed-vertex structure.

4 Six points: the hexagon case

The same technique works at $n = 6$, with two new features:

- The hexagon has *two classes* of chords – short diagonals (class 1) and long diagonals (class 2).
- Each zero now has *three* substitutions, of which *two* are non-bare. This produces an additional order-4 layer of kills coming from monomials with two non-bare substituted indices.

4.1 Setup at $n = 6$

The hexagon has 9 chords:

Short diagonals (class 1, enclose 1 vertex): $X_{13}, X_{24}, X_{35}, X_{46}, X_{15}, X_{26}$
Long diagonals (class 2, enclose 2 vertices): X_{14}, X_{25}, X_{36}

Using cyclic shorthand,

$$x_1 = X_{13},\ x_2 = X_{24},\ x_3 = X_{35},\ x_4 = X_{46},\ x_5 = X_{15},\ x_6 = X_{26}; \quad y_1 = X_{14},\ y_2 = X_{25},\ y_3 = X_{36}.$$

The enclosed-vertex sets are

$$E(x_r) = \{r+1\} \bmod 6, \qquad E(y_r) = \{r+1, r+2\} \bmod 6.$$

Ansatz. Each monomial has $n - 3 = 3$ chord factors. The number of multisets of size 3 from 9 chords is $\binom{9+2}{3} = 165$. Of these 14 are tree triples (the $C_4 = 14$ triangulations of the hexagon), and 151 are non-local.

4.2 The six zeros, derived by hand

Each $\mathcal{Z}_r$ is defined by $c_{r,r+2} = c_{r,r+3} = c_{r,r+4} = 0$ (three constraints). For $\mathcal{Z}_1$, the mesh identity gives

$$\begin{aligned}
c_{1,3} &= X_{14} + X_{23} - X_{13} - X_{24} = y_1 - x_1 - x_2 = 0 \implies x_2 = y_1 - x_1,\\
c_{1,4} &= X_{15} + X_{24} - X_{14} - X_{25} = x_5 + x_2 - y_1 - y_2 = 0 \implies y_2 = x_5 - x_1,\\
c_{1,5} &= X_{16} + X_{25} - X_{15} - X_{26} = 0 + y_2 - x_5 - x_6 = 0 \implies x_6 = -x_1.
\end{aligned}$$

So for $\mathcal{Z}_1$, the special variable is $x_* = x_1 = X_{13}$, and the three substituted chords $\{x_2, y_2, x_6\}$ are exactly the chords from the enclosed vertex $E(x_1) = \{2\}$. By cyclic rotation:

Zone	Substitutions			Free set F_r (excluding special)
$\mathcal{Z}_1$	$x_2 = y_1 - x_1$	$y_2 = x_5 - x_1$	$x_6 = -x_1$	$\{x_3, x_4, x_5, y_1, y_3\}$
$\mathcal{Z}_2$	$x_3 = y_2 - x_2$	$y_3 = x_6 - x_2$	$x_1 = -x_2$	$\{x_4, x_5, x_6, y_1, y_2\}$
$\mathcal{Z}_3$	$x_4 = y_3 - x_3$	$y_1 = x_1 - x_3$	$x_2 = -x_3$	$\{x_1, x_5, x_6, y_2, y_3\}$
$\mathcal{Z}_4$	$x_5 = y_1 - x_4$	$y_2 = x_2 - x_4$	$x_3 = -x_4$	$\{x_1, x_2, x_6, y_1, y_3\}$
$\mathcal{Z}_5$	$x_6 = y_2 - x_5$	$y_3 = x_3 - x_5$	$x_4 = -x_5$	$\{x_1, x_2, x_3, y_1, y_2\}$
$\mathcal{Z}_6$	$x_1 = y_3 - x_6$	$y_1 = x_4 - x_6$	$x_5 = -x_6$	$\{x_2, x_3, x_4, y_2, y_3\}$

In every zone: one special x_* (a short diagonal), three substituted chords from its enclosed vertex (two non-bare with companions in F_r; one bare $= -x_*$), and five free chords (3 short, 2 long).

4.3 The kill technique at $\mathcal{Z}_1$: three layers

For $\mathcal{Z}_1$: special $x_* = x_1$; non-bare substitutions $x_2 = y_1 - x_1$ (companion $f_{x_2} = y_1$) and $y_2 = x_5 - x_1$ (companion $f_{y_2} = x_5$); bare $x_6 = -x_1$; free set $F_1 = \{x_3, x_4, x_5, y_1, y_3\}$ (five chords).

The relevant Laurent expansions are

$$\frac{1}{x_2} = -\frac{1}{x_1} - \frac{y_1}{x_1^2} - \frac{y_1^2}{x_1^3} - \cdots, \qquad \frac{1}{y_2} = -\frac{1}{x_1} - \frac{x_5}{x_1^2} - \frac{x_5^2}{x_1^3} - \cdots, \qquad \frac{1}{x_6} = -\frac{1}{x_1}.$$

Layer 1: order x_1^0 (limit $x_1 \to \infty$). [35 kills] As $x_1 \to \infty$, all of x_2, y_2, x_6 tend to $\pm\infty$, so $1/x_2, 1/y_2, 1/x_6 \to 0$. The surviving monomials of B are those with all three indices in F_1:

$$\big(B|_{\mathcal{Z}_1}\big)\big|_{x_1\to\infty} \;=\; \sum_{(a,b,c)\subseteq F_1} \frac{a_{abc}}{X_a X_b X_c}.$$

The number of multisets (a,b,c) of size 3 from 5 elements is $\binom{5+2}{3} = 35$. Each $1/(X_aX_bX_c)$ is a distinct rational monomial in the five free variables; they are linearly independent. Hence each $a_{abc} = 0$:

$$\boxed{\text{35 single-term kills at order 0: all } a_{abc} \text{ with } \{X_a, X_b, X_c\} \subseteq F_1.}$$

Layer 2: order x_1^{-2} (coefficient of $1/x_1^2$). [20 kills] Consider monomials of B with exactly one factor among the non-bare substituted chords $\{x_2, y_2\}$, the other two factors in F_1. We argue by the "unique-source" principle.

Sub-case A: monomial $a_{x_2X_aX_b}/(x_2X_aX_b)$ with $X_a, X_b \in F_1$. Substituting $1/x_2 = -1/x_1 - y_1/x_1^2 - \cdots$, the $1/x_1^2$ piece is $-y_1/(x_1^2X_aX_b)$, contributing

$$-\frac{a_{x_2X_aX_b}\, y_1}{X_aX_b} \quad \text{to the } 1/x_1^2 \text{ coefficient.}$$

When is the free-variable monomial $y_1/(X_aX_b)$ a unique source? The only other potential contribution would come from $1/y_2$'s tail, but that tail is x_5/x_1^2, not y_1. So $y_1/(X_aX_b)$ uniquely identifies the $a_{x_2X_aX_b}$ origin *provided* neither X_a nor X_b equals y_1 (otherwise the y_1 in the numerator cancels with the $1/y_1$ in the denominator, producing a degenerate monomial that mixes with other contributions).

Hence the unique-source condition is $\{X_a, X_b\} \subseteq F_1 \setminus \{y_1\} = \{x_3, x_4, x_5, y_3\}$. The number of such multisets of size 2 from 4 elements is $\binom{4+1}{2} = 10$:

$$\boxed{\text{10 single-term kills at order 2: all } a_{x_2X_aX_b} = 0 \text{ with } \{X_a, X_b\} \subseteq \{x_3, x_4, x_5, y_3\}.}$$

Sub-case B: monomial $a_{y_2X_aX_b}/(y_2X_aX_b)$ with $X_a, X_b \in F_1$. Same argument with y_2's companion x_5 instead of y_1's. Unique-source condition: $\{X_a, X_b\} \subseteq F_1 \setminus \{x_5\} = \{x_3, x_4, y_1, y_3\}$. Another $\binom{4+1}{2} = 10$ kills.

(Sub-case for x_6: bare substitution, contributes 0 to the order-2 companion structure since $1/x_6$ has no tail.)

Layer 3: order x_1^{-4} (coefficient of $1/x_1^4$). [3 new kills] Consider monomials of B with *both* non-bare substitutions as factors: $a_{x_2y_2X_c}/(x_2y_2X_c)$ with $X_c \in F_1$. Use the product Laurent expansion

$$\frac{1}{x_2y_2} \;=\; \frac{1}{(y_1-x_1)(x_5-x_1)} \;=\; \frac{1}{x_1^2}\cdot\frac{1}{(1-y_1/x_1)(1-x_5/x_1)}.$$

Expanding in $1/x_1$,

$$\frac{1}{(1-y_1/x_1)(1-x_5/x_1)} \;=\; 1 + \frac{y_1+x_5}{x_1} + \frac{y_1^2+y_1x_5+x_5^2}{x_1^2} + \mathcal{O}(x_1^{-3}),$$

so $\left.\dfrac{1}{x_2y_2}\right|_{1/x_1^4} = \dfrac{y_1^2+y_1x_5+x_5^2}{x_1^4}$. The coefficient at $1/x_1^4$ from $a_{x_2y_2X_c}/(x_2y_2X_c)$ is therefore

$$\frac{a_{x_2y_2X_c}\,(y_1^2+y_1x_5+x_5^2)}{X_c}.$$

The mixed monomial y_1x_5/X_c is a unique source for the coefficient $a_{x_2y_2X_c}$, provided $X_c \neq y_1, x_5$ (else degenerate). Hence $X_c \in F_1 \setminus \{y_1, x_5\} = \{x_3, x_4, y_3\}$:

$$\boxed{\text{3 single-term kills at order 4: } a_{x_2y_2x_3} \;=\; a_{x_2y_2x_4} \;=\; a_{x_2y_2y_3} \;=\; 0.}$$

Higher orders contribute no new single-term kills. Monomials with three non-bare/bare-factor combinations either involve x_6 (bare, no tail) or repeat the same Laurent structure already captured. Direct expansion shows no new isolable single-term equations.

Total kills from $\mathcal{Z}_1$: $35 + 20 + 3 = 58$ single-coefficient kills.

4.4 Cycling: union over six zones is 151

By cyclic symmetry every $\mathcal{Z}_r$ produces $35 + 20 + 3 = 58$ kills, with the same internal structure just rotated. Tabulating all $6 \times 58 = 348$ raw kills and accounting for overlaps yields a union of exactly 151 distinct coefficients – exactly the number of non-local multisets at $n = 6$.

Sketch of the union argument. It suffices to show every non-local multiset is killed by at least one zone. The non-local multisets at $n = 6$ split by chord-class composition (#short, #long):

Composition	# multisets	# triangulations among them
3 short	$\binom{6+2}{3} = 56$	2 (the two "zigzag" triangulations $x_1x_3x_5,\ x_2x_4x_6$)
2 short, 1 long	$3 \cdot \binom{6+1}{2} = 63$	6 (six fan-like)
1 short, 2 long	$6 \cdot \binom{3+1}{2} = 36$	6
3 long	$\binom{3+2}{3} = 10$	0
Total	165	14

For each non-tree multiset, one identifies a zone $\mathcal{Z}_r$ where the multiset's indices fit one of the three layers (all in F_r; exactly one non-bare substituted index with the rest in $F_r \setminus \{f_s\}$; or both non-bare substituted indices with the third in $F_r \setminus \{f_{s_1}, f_{s_2}\}$). A full case analysis confirms every non-tree multiset is caught.

The 14 tree triples are not killed: each is a triangulation (non-crossing chords), so for every zone the multiset includes either the special x_* itself, or a chord that violates the unique-source conditions of the three layers.

Theorem 2 (Locality at six points)**.** *The six hidden one-zeros of the hexagon force every non-local coefficient of the general rational ansatz* (2) *to vanish. The surviving* 14 *coefficients are exactly the tree triangulations of the hexagon.*

5 Looking ahead: the structural pattern

The kill structure at $n = 4, 5, 6$ is governed by three numbers per zone:

- $|F_r|$, the number of free chords (excluding the special). At n points: $|F_r| = \dfrac{n(n-3)}{2} - (n-3) - 1$.
- $n - 4$, the number of *non-bare* substituted chords.
- $n - 3$, the multiset size (chord factors per monomial of B).

The general per-zone kill count, verified at $n = 4, 5, 6$, is the sum over layers, where layer k corresponds to multisets with exactly k non-bare substituted indices:

$$\text{kills per zone} = \sum_{k=0}^{\lfloor (n-3)/2 \rfloor} \binom{n-4}{k} \cdot \binom{|F_r| - k + (n-3-k) - 1}{n-3-k}. \tag{5}$$

Plugging in:

- $n = 4$: $|F_r| = 0$; sum $= 0$ (no non-local kills, as expected).
- $n = 5$: $|F_r| = 2$; sum $= 3 + 1 = 4$ kills/zone, union $= 10$.
- $n = 6$: $|F_r| = 5$; sum $= 35 + 20 + 3 = 58$ kills/zone, union $= 151$.

This is a clean recursive template. The general-n proof should proceed along the same lines: derive each zero by hand using the mesh identity; identify the special variable via the enclosed-vertex structure; and verify layer by layer that the unique-source isolation of single-coefficient kills covers all non-local multisets. The class hierarchy of chords (short, long, extra-long, etc.) and the palindromic class structure of the substitutions suggest an inductive argument that we leave for future work.